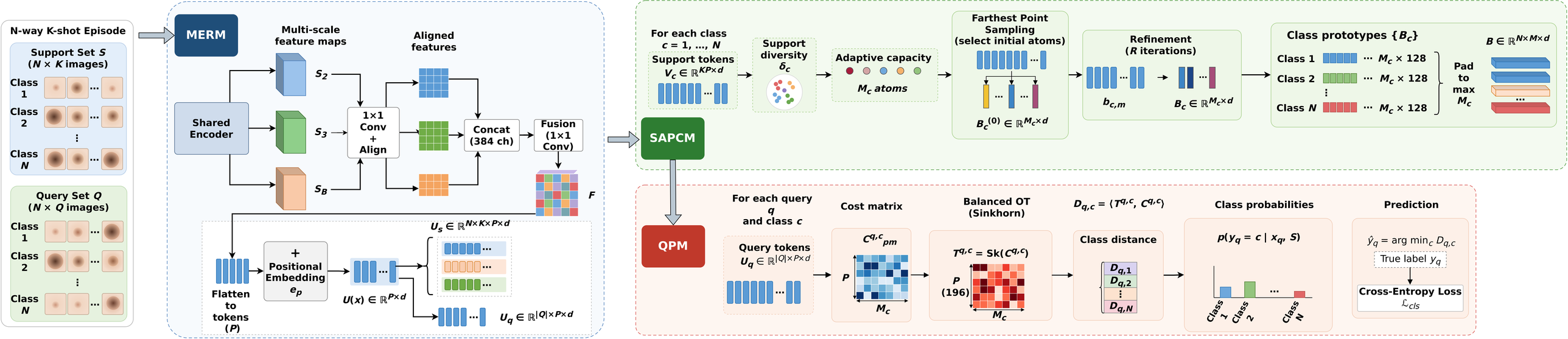

N-way K-shot Episode
Support Set S (N × K images)
Class 1
Class 2
Class N
Query Set Q (N × Q images)
Class 1
Class 2
Class N
MERM
Multi-scale feature maps
Aligned features
Shared Encoder
S2
S3
SB
1×1 Conv + Align
Concat (384 ch)
Fusion (1×1 Conv)
F
Us ∈ ℝ^{N×K×P×d}
Flatten to tokens (P)
+ Positional Embedding ep
U(x) ∈ ℝ^{P×d}
Uq ∈ ℝ^{|Q|×P×d}
SAPCM
For each class c = 1, ..., N
Support tokens Vc ∈ ℝ^{KP×d}
Support diversity δc
Adaptive capacity
Mc atoms
Farthest Point Sampling (select initial atoms)
Bc(0) ∈ ℝ^{Mc×d}
Refinement (R iterations)
bc,m
Bc ∈ ℝ^{Mc×d}
Class prototypes {Bc}
Class 1 ··· Mc × 128
Class 2 ··· Mc × 128
Class N ··· Mc × 128
Pad to max Mc
B ∈ ℝ^{N×M×d}
QPM
For each query q and class c
Query tokens Uq ∈ ℝ^{|Q|×P×d}
Cost matrix
C^{q,c}pm
P
Mc
Balanced OT (Sinkhorn)
T^{q,c} = Sk(C^{q,c})
P (196)
Mc
Dq,c = ⟨T^{q,c}, C^{q,c}⟩
Class distance
Dq,1
Dq,2
Dq,N
Class probabilities
p(yq = c | xq, S)
Class 1
Class 2
Class N
Prediction
ŷq = arg min_c Dq,c
True label yq
Cross-Entropy Loss
ℒcls
: Feature token (d)
: Forward flow
: Backward flow (gradient)
: Matrix (darker = higher value)
MERM: Multi-scale Evidence Representation Module
SAPCM: Support-diversity Adaptive Prototype Construction Module
QPM: Query-to-Prototype Matching Module